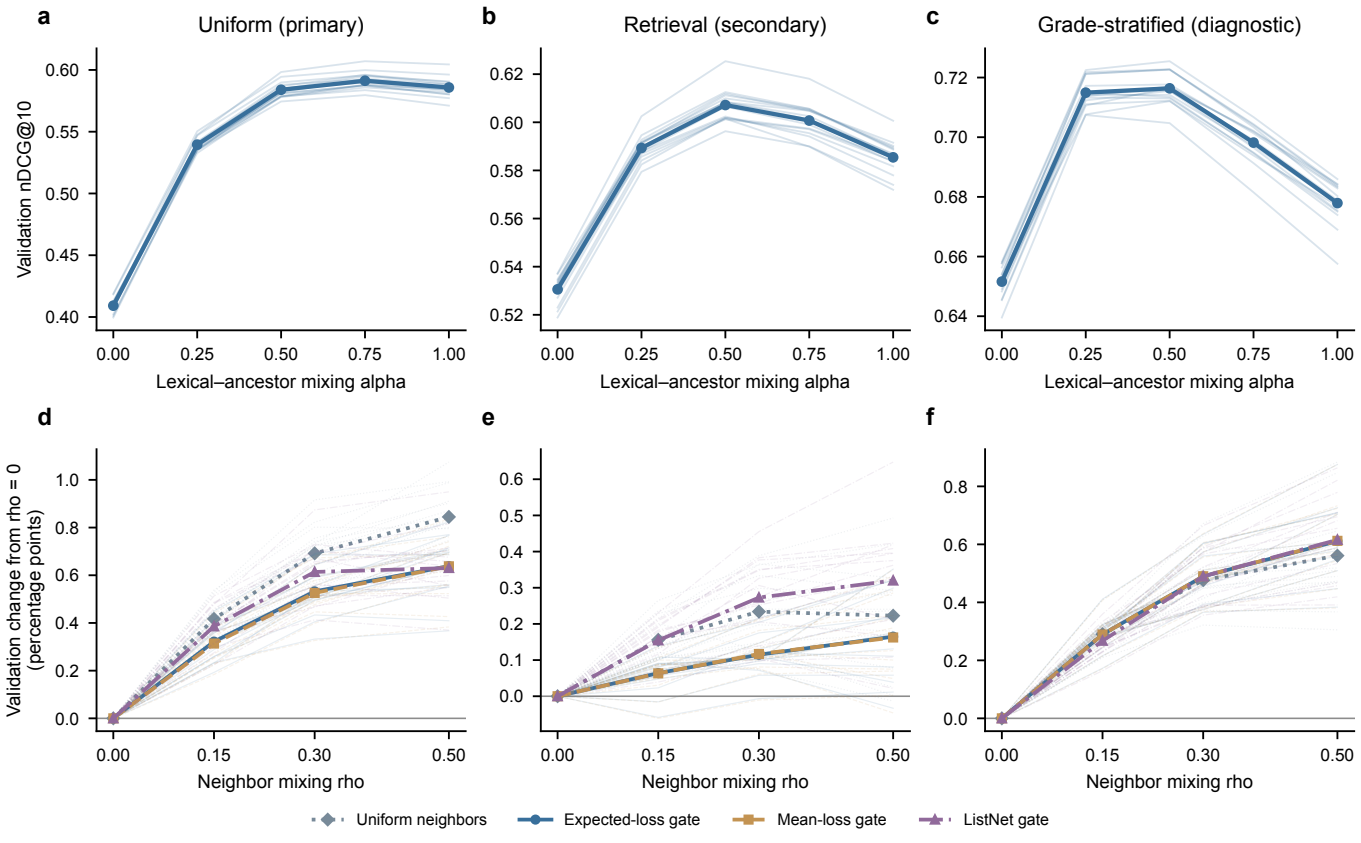
a
Uniform (primary)
b
Retrieval (secondary)
c
Grade-stratified (diagnostic)
Validation nDCG@10
0.60
0.55
0.50
0.45
0.40
0.62
0.60
0.58
0.56
0.54
0.52
0.72
0.70
0.68
0.66
0.64
0.00
0.25
0.50
0.75
1.00
Lexical–ancestor mixing alpha
d
e
f
Validation change from rho = 0
(percentage points)
1.0
0.8
0.6
0.4
0.2
0.0
0.5
0.3
0.1
0.15
0.30
0.50
Neighbor mixing rho
Uniform neighbors
Expected-loss gate
Mean-loss gate
ListNet gate